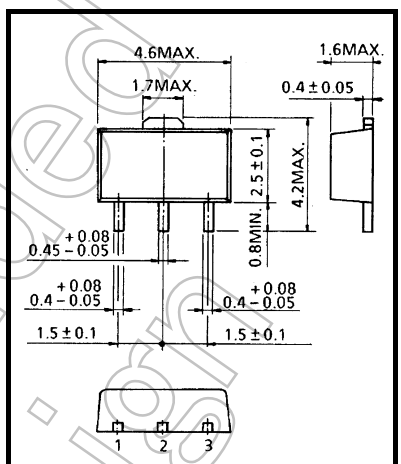
TOSHIBA Transistor Silicon PNP Epitaxial Type (PCT process)

2SA1736

Power Amplifier Applications
Power Switching Applications

- Low saturation voltage: $V_{CE\ (sat)}$ = −0.5 V (max) (I_C = −1.5 A)
- High speed switching time: t_{stg} = 0.2 μs (typ.)
- Small flat package
- P_C = 1.0 to 2.0 W (mounted on a ceramic substrate)
- Complementary to 2SC4541

Unit: mm



1. Base
2. Collector (heat sink)
3. Emitter

PW-MINI	
JEDEC	—
JEITA	SC-62
TOSHIBA	2-5K1A

Weight: 0.05 g (typ.)

Absolute Maximum Ratings (Ta = 25°C)

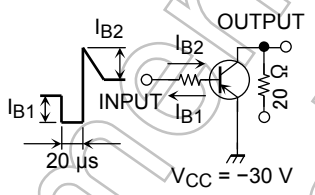
Characteristics	Symbol	Rating	Unit
Collector-base voltage	V_{CBO}	−60	V
Collector-emitter voltage	V_{CEO}	−50	V
Emitter-base voltage	V_{EBO}	−6	V
Collector current	I_C	−3	A
Base current	I_B	−0.6	A
Collector power dissipation	P_C	500	mW
	P_C (Note 1)	1000	
Junction temperature	T_j	150	°C
Storage temperature range	T_{stg}	−55 to 150	°C

Note 1: Mounted on a ceramic substrate (250 mm^2 × 0.8 t)

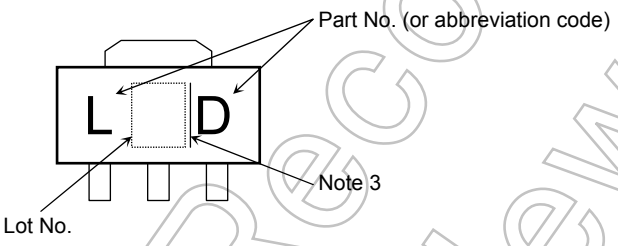
Note 2: Using continuously under heavy loads (e.g. the application of high temperature/current/voltage and the significant change in temperature, etc.) may cause this product to decrease in the reliability significantly even if the operating conditions (i.e. operating temperature/current/voltage, etc.) are within the absolute maximum ratings.
Please design the appropriate reliability upon reviewing the Toshiba Semiconductor Reliability Handbook (“Handling Precautions”/Derating Concept and Methods) and individual reliability data (i.e. reliability test report and estimated failure rate, etc).

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Electrical Characteristics (Ta = 25°C)

Characteristics		Symbol	Test Condition	Min	Typ.	Max	Unit
Collector cut-off current		I_{CBO}	V_{CB} = −60 V, I_E = 0	—	—	−0.1	μA
Emitter cut-off current		I_{EBO}	V_{EB} = −6 V, I_C = 0	—	—	−0.1	μA
Collector-emitter breakdown voltage		$V_{(BR)\ CEO}$	I_C = −10 mA, I_B = 0	−50	—	—	V
DC current gain		$h_{FE\ (1)}$	V_{CE} = −2 V, I_C = −100 mA	120	—	400	
		$h_{FE\ (2)}$	V_{CE} = −2 V, I_C = −2 A	40	—	—	
Collector-emitter saturation voltage		$V_{CE\ (sat)}$	I_C = −1.5 A, I_B = −75 mA	—	—	−0.5	V
Base-emitter saturation voltage		$V_{BE\ (sat)}$	I_C = −1.5 A, I_B = −75 mA	—	—	−1.2	V
Transition frequency		f_T	V_{CE} = −2 V, I_C = −100 mA	—	100	—	MHz
Collector output capacitance		C_{ob}	V_{CB} = −10 V, I_E = 0, f = 1 MHz	—	32	—	pF
Switching time	Turn-on time	t_{on}		—	0.1	—	μs
	Storage time	t_{stg}		—	0.2	—	
	Fall time	t_f	I_{B1} = 75 mA, I_{B2} = 75 mA, DUTY CYCLE ≤ 1%	—	0.1	—	

Marking

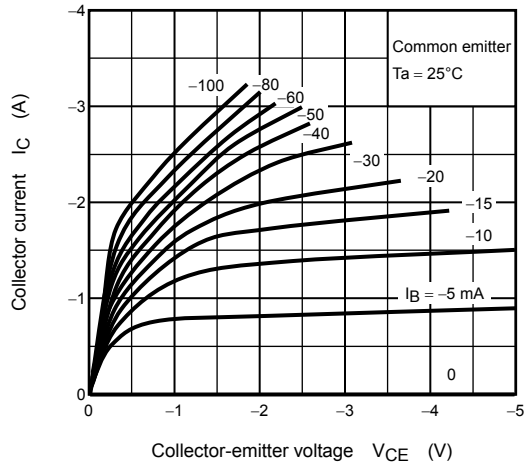


Note 3: A line beside a Lot No. identifies the indication of product Labels.
Without a line: [[Pb]]/INCLUDES > MCV
With a line: [[G]]/RoHS COMPATIBLE or [[G]]/RoHS [[Pb]]

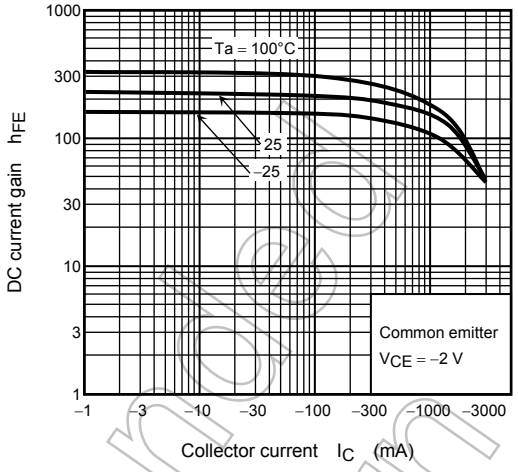
Please contact your TOSHIBA sales representative for details as to environmental matters such as the RoHS compatibility of Product. The RoHS is the Directive 2002/95/EC of the European Parliament and of the Council of 27 January 2003 on the restriction of the use of certain hazardous substances in electrical and electronic equipment.

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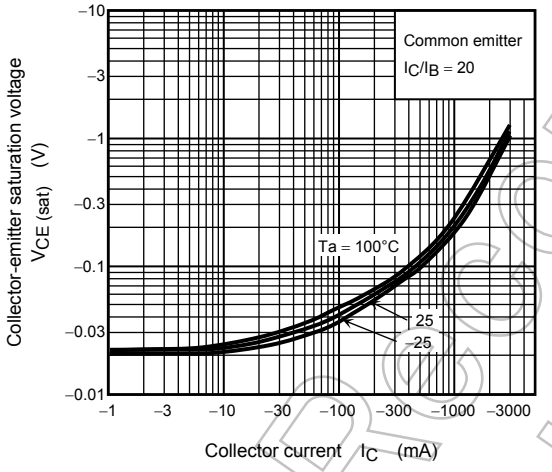
$I_C - V_{CE}$



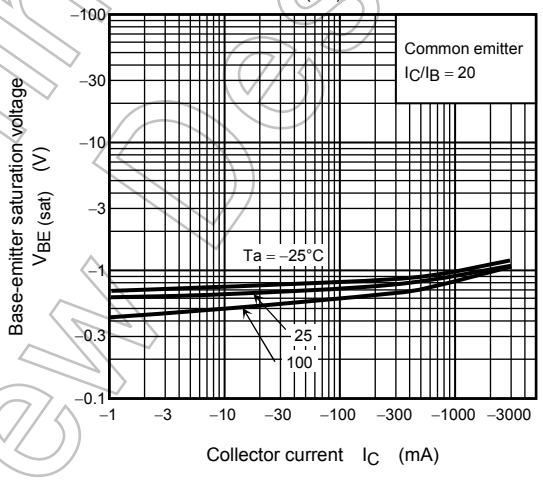
$h_{FE} - I_C$



$V_{CE\ (sat)} - I_C$

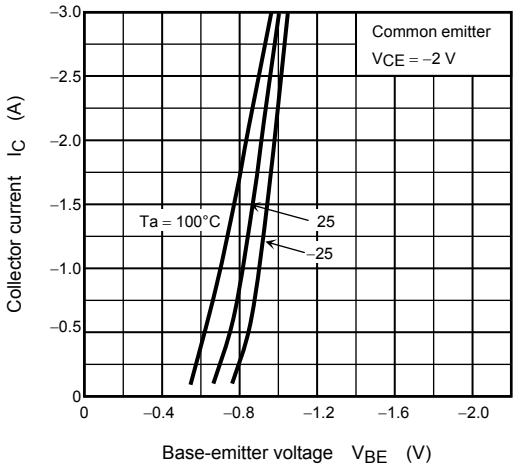


$V_{BE\ (sat)} - I_C$

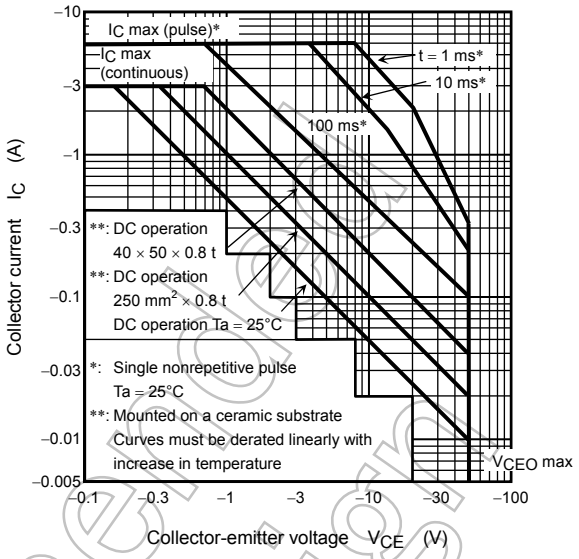


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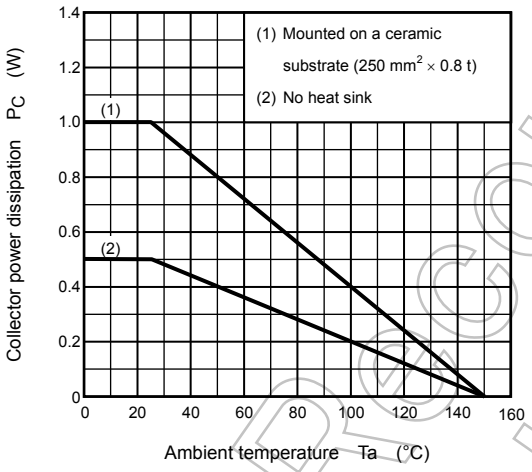
$I_C - V_{BE}$



Safe Operating Area



$P_C - Ta$



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